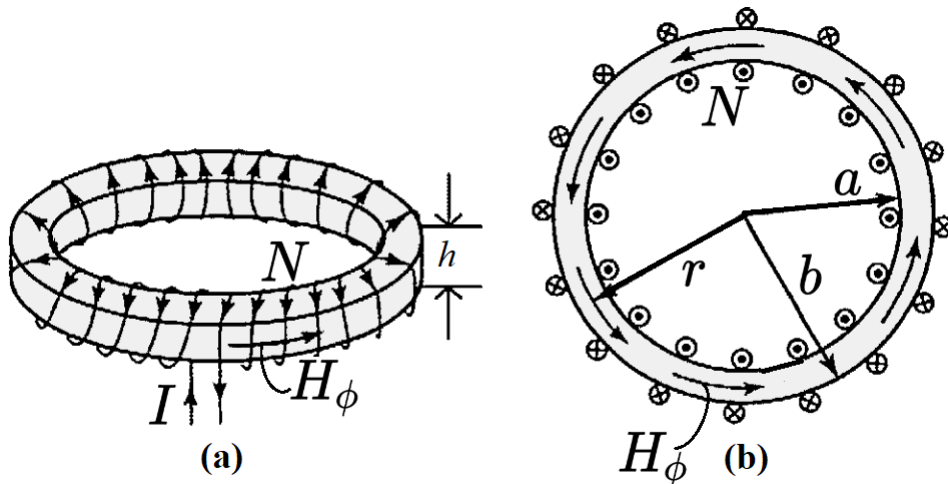


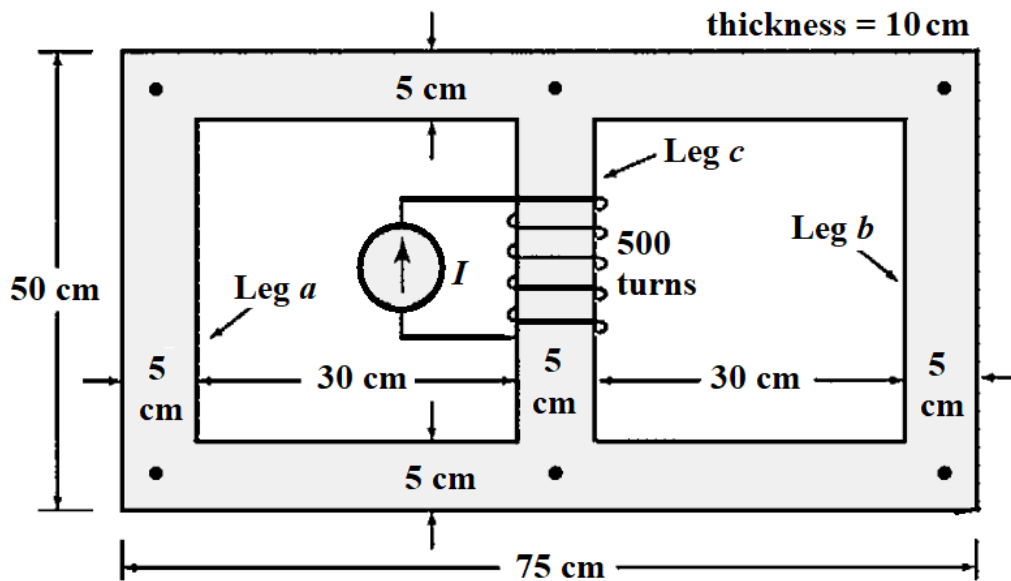


Q1: [100] A closely spaced winding with N turns is wound in the form of a ring, as shown in Figure 1. The inner and the outer radii of the ring are a and b , respectively. The height of the ring is h . If the winding carries a current of I amperes, find

- (a) the magnetic field intensity within the ring,
- (b) the magnetic flux density,
- (c) the total flux enclosed by the ring,
- (d) the inductance, and
- (e) the total stored energy.



Q2: [50] If the 500-turn coil in Figure 2 carries a current of 2 A, determine the flux in each leg of the magnetic circuit. The relative permeability of the core is 1000.



Q3: [50] Find the per unit length inductance of the following three-wire circuit:

